

ANSWERS

1.

Question	Answer	Marks
1(a)(i)	particles move / collide / hit	B1
	particles collide with (inner) walls (of rocket)	B1
	force exerted by collisions over / on an area	B1
1(a)(ii)	more collisions per unit time	B1

2.

Question	Answer	Marks
4(a)(i)	$\Delta p = h\rho g$ or $1.0 \times 1.4 \times 10^4 \times 9.8$	C1
	1.4×10^5 or 1.372×10^5 or $1.0 \times 10^5 + (1.0 \times 1.4 \times 10^4 \times 9.8)$ or $1.0 \times 10^5 + 1.4 \times 10^5$	C1
	2.4×10^5 (Pa)	A1
4(a)(ii)	$(F =) pA$ or $2.4 \times 10^5 \times 4.0 \times 10^{-4}$	C1
	95 (N) or 96 (N)	A1
4(b)(i)	the liquid falls (in the tube)	B1
	then stops falling or reaches stability	B1
	(original) pressure of liquid at bottom of tube greater than pressure of liquid in basin immediately beneath it or pressure (of liquid in tube) becomes equal to atmospheric pressure / pressure in the container or leaving a vacuum (at the closed end of the tube)	B1
4(b)(ii)	height of liquid surface (in tube measured)	B1
	measured relative to level of liquid surface in container or $h\rho g$ (used) to calculate the pressure	B1

3.

Question	Answer	Marks
4(a)(i)	$(P =) \text{ Force / Area}$ algebraic or numerical	C1
	520 000 (Pa)	A1
4(a)(ii)	They are equal (in size and direction)	B1
4(a)(iii)	small(er) area of contact (with board)	C1
	large(r) pressure (on board)	A1
4(b)(i)	<u>pressure</u> (of water) act / <u>pressure</u> in bag	B1
	<u>force</u> (on water) at right angles to the surface / in a straight line or pressure acts in all directions	B1
4(b)(ii)	larger pressure at A / greater weight of water above A (and greater height / depth / mass / weight (of water above B)	B1

4.

Question	Answer	Marks
3(a)(i)	<u>mercury</u> exerts a changing or reduced pressure / force (when vertical but not when horizontal) / <u>mercury</u> stops exerting a pressure	B1
3(a)(ii)	particles hit walls / mercury	B1
	fewer hits per second or hit less often / less frequently or fewer hits per unit area	B1
	less force (per unit area on wall)	B1
3(b)	$(P =) \rho gh$ in any form	C1
	41 000 seen	C1
	140 000 (Pa)	A1
3(c)	curved shape with decreasing gradient	B1
	points plotted at or curve through $(\frac{1}{2}V_0, 2P_0)$, (P_0, V_0) and $(2V_0, \frac{1}{2}P_0)$	B1

5.

Question	Answer	Marks
3(a)	force divided by area	B1
3(b)	(molecules) move (around at random) / have K.E.	B1
	(molecules) hit sides / piston	B1
	(molecules) cause force on piston / sides	B1
3(c)(i)	$10 / 1.2 \times 10^{-4}$ or 8.3×10^4 (Pa) seen	C1
	1.8×10^5 Pa	A1
3(c)(ii)	$P_1 V_1 = P_2 V_2$ algebraic or numerical in any form or $P \propto 1 / V$	B1
	27 or 28 cm ³	B1

6.

Question	Answer	Marks
8(a)	(normal) force per unit area or (normal) force $\div$ area	B1
8(b)(i)	$(p =) \rho gh$ or $25 \times 1000 \times 10$	C1
	2.5×10^5 Pa	A1
8(b)(ii)	1.0×10^5 Pa	B1
8(c)(i)	pressure increases with depth	B1
	pressure / force on outer face greater than that on inner face	B1
	resultant force on piston	B1
8(c)(ii)	pressure increases and more collisions (of molecules with walls of cylinder)	B1
	more frequent collisions or more collisions per unit area	B1
	molecules have less distance to travel between collisions (with walls) or molecular density greater	B1
8(c)(iii)	<u>curve</u> with negative gradient and gradient of decreasing magnitude	B1
	<u>negative gradient</u> at V_0 and approaches x-axis asymptotically	B1
8(c)(iv)	atmospheric pressure varies or temperature varies	B1
8(c)(v)	molecules touching or (very) close together (and cannot be forced closer)	B1
	(repulsive) forces between molecules (very) large	B1

7.

Question	Answer	Marks
1(a)(i)	pressure due to mercury (in tube) is greater than pressure due to atmosphere (+ small height of mercury)	B1
	(resultant) force downwards and mercury flows out of tube	B1
1(a)(ii)	vacuum or nothing or mercury vapour	B1
1(b)	height of mercury column mentioned	C1
	height of mercury column <u>above level in dish measured</u>	A1
	candidate suggests use of $p = h \rho g$ and h in metres (m)	B1

8.

Question	Answer	Marks
2(a)(i)	$(F =) pA$ or $3.8 \times 10^5 \times 6.1 \times 10^{-4}$	C1
	230 N	A1
2(a)(ii)	force on other side of piston due to air / atmospheric pressure or pressure of trapped air decreases as valve opens (and air enters the tyre)	B1
2(b)	molecules collide <u>with walls / piston / pump / surface</u>	B1
	number of molecules per unit volume increases (as volume decreases)	B1
	more collisions per unit area or more frequent collisions / collide more often	B1

9.

Question	Answer	Marks
2(a)(i)	$(p =) \rho gh$ or $940 \times 10 \times 3.3$	C1
	3.1×10^4 Pa	A1
2(a)(ii)	atmospheric pressure acts at both ends of crack	B1
2(b)	(surface) level decreases / drops	B1
	pressure (at level of crack) decreases	B1

10.

Question	Answer	Marks
2(a)(i)	(pressure =)force / area	B1
2(a)(ii)	fewer molecules or less gas or more space / further apart	B1
	less frequent collisions (with walls)	B1
	less force exerted on walls	B1

11.

- (a) $(p =)h\rho g$ **or** $32 \times 1000 \times 10$ C1
 3.2×10^5 Pa A1
- (b) (i) atmospheric pressure (is also acting on the surface of the water) B1
- (ii) $(F =)pA$ **or** $4.2 \times 10^5 \times 45$ **or** $3.2 \times 10^5 \times 45$ **or** 1.44×10^7 C1
 $1.9 / 1.89 \times 10^7$ N A1

- 12.**
- (a) tape B1
- (b) (i) mass $\div$ volume **or** mass per unit volume B1
- (ii) (V=) $15 \times 0.25 \times 2$ or 7.5 seen C1
 2400 kg/m^3 A1
- (iii) (A=) 15×0.25 **or** 3.75 **or** (P=) F/A C1
or (P=) $180000/A$ **or** (P=) $dgh / \rho gh$ seen A1
48000 Pa
- (iv) (length doubles) so both area **and** weight/force double B1
or area **and** force/weight both increase/larger (in proportion)
or height **and** density the same (in $P = dgh$)
- 13.**
- (a) (i) force per unit area **or** force divided by area **or** force on 1 m^2 **or** force on unit area [B1]
- (ii) more water above **or** more force from water **or** more atoms/molecules above [C1]
larger weight of water above [A1]
- (iii) Pa **or** N/m^2 [B1]
- 14.**
- (a) Q and R B1
- (b) ρgh in any form, algebraic or numerical C1
 $1.0(336) \times 10^5 \text{ N/m}^2$ A1
- (c) water is less dense **or** has density $1000 \text{ (kg/m}^3\text{)}$ B1
water further up tube/fills tube **or** height greater **or** water enters pump **or** water boils B1

- 15.**
- (a) 14 N B1
- (b) ($P =$) F/A algebraically in symbols or words in any form; $14/3.0 \times 10^{-5}$ C1
 $4.67 \times 10^5 \text{ Pa}$; $4.7 \times 10^5 \text{ Pa}$ ecf (a) A1
- (c) stiffer/stronger spring; piston has less area/diameter; smaller piston (and tube) B1
- (d) molecules/particles/atoms collide with tyre/walls/piston B1
 air/molecules enter gauge; fewer molecules in the tyre; fewer hits/sec;
 less frequent hits; volume increases B1
- 16.**
- (a) (i) weight of water (causes extra pressure)(**not** mass) B1
- (ii) density of liquid/(sea-)water **or** gravitational field strength/acceleration of freefall
 (**not** gravity) B1
- (b) (i) $3.6/3.60 \times 10^5 \text{ Pa}$ B1
- (ii) $P_1 V_1 = P_2 V_2$ **or** $1.05 \times 10^5 \times 6000 = 3.60 \times 10^5 \times V_2$ C1
 1700 **or** 1750 **or** 1800 cm^3 A1